

GLHS: A Co-Versioned Disclosure-to-Commit Governance Contract for Longitudinal Health AI

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Abstract

Persistent health AI creates a consistency problem that retrieval quality alone cannot solve. A model may read an authorized view of a longitudinal record, reason over it, and return a persistent proposal only after the record, consent state, authorization, or governing policy has changed. GLHS treats this read-to-write interval as a governance boundary. Its central mechanism is a disclosure-conditioned, co-versioned contract: on the snapshot-bound path, the exact Task-Bounded Health State Snapshot (THSS) actually supplied to the model becomes an auditable dependency of the later persistent proposal, and a Governed State Transition (GST) revalidates the relevant state and governance coordinates before commit. The implementation preserves bitemporal state, provenance, disclosure scope, snapshot identity and digests, policy and consent versions, and optimistic-concurrency coordinates, while avoiding novelty claims for those established components individually. We evaluate the contract through deterministic conformance tests, PostgreSQL concurrency and service-overhead experiments, governance-writer TOCTOU schedules, bounded finite-state exploration, and model-context studies. Snapshot-bound proposals failed closed on the tested identity, digest, evidence, state, policy, consent, purpose, role, task, and expiry mismatches. In a frozen 12-schedule PostgreSQL governance-TOCTOU matrix, no forbidden commit or indeterminate ordering was observed across consent revocation, role change, policy-epoch advance, state change, and compound-drift schedules. Bounded exploration to depth 5 covered 21,361 states and 90,432 transitions with no violation of 11 specified invariants. Model-context experiments were deliberately secondary: a frozen 64-subject cohort reached near-ceiling Strict-THSS accuracy, whereas a later 384-subject Claude-only comparison with full authorized history was null (70 wins, 73 losses, 241 ties; mean effect -0.0078 , 95% CI $[-0.0677, 0.0521]$, $p = 0.867$). We therefore treat these runs as context-utility evidence rather than evidence that GLHS makes models uniformly more accurate. Overall, the study supports a traceable software path from governed disclosure to persistent write under the tested conditions; it does not establish clinical effectiveness, correctness for arbitrary unbounded interleavings, or regulatory compliance.

Keywords: longitudinal health AI; stateful AI; disclosure-conditioned commit; health-data governance; bitemporal state; snapshot binding; provenance; consent

1 Introduction

A longitudinal health record is not simply a collection of documents to retrieve. The same fact can have one clinical effective time and a later knowledge time; sources can disagree; and information that is appropriate for one actor, purpose, or task may be inappropriate for another. A health-AI system can

therefore retrieve a relevant item and still present the wrong historical state, disclose more information than the current use requires, or act on context that is no longer valid by the time a proposed change is written back.

The underlying building blocks are well established. Medical database research distinguishes valid time from transaction or knowledge time [1, 2, 3]. FHIR provides Provenance, Consent, and version-aware HTTP primitives [4, 5, 6]; openEHR maintains versioned and reconstructable EHR contributions [7]; and W3C PROV provides a general lineage model [8]. Recent health-AI systems preserve temporal conflicts, reconcile patient and EHR streams, and maintain longitudinal memory [9, 10, 11, 12, 16]. Agent-memory and tool-runtime research now also supports transactional updates, recovery, scoped propagation, bitemporal write semantics, and staged irreversible effects [13, 14, 15, 26, 25]. Work on concurrent agents has moved even closer to the read–write boundary: verified runtimes formalize read–generate–write anomalies [27], commit-time authorization requires authority evidence to remain valid at durability [28], and policy-state serializability rechecks authorization against current policy state at commit [29].

These advances make the remaining question narrower, not broader. GLHS does not claim that transactional commit, bitemporality, consent, provenance, optimistic concurrency, or commit-time authorization is new. The specific problem studied here is whether the *exact governed health disclosure actually consumed during inference* can remain an explicit, auditable dependency of a later persistent proposal. A base-state or generic authority check can establish freshness without identifying the health content and governance coordinates that supplied the model’s evidence. Conversely, a carefully minimized read-time context can disappear from the write-admission path unless the two operations are deliberately linked.

GLHS addresses that gap with a disclosure-conditioned, co-versioned read–write contract. A Task-Bounded Health State Snapshot (THSS) records the governed health state shown to the model together with its subject, actor, purpose, task, state, policy, consent, provenance, expiry, and digest coordinates. On the evaluated snapshot-bound path, a persistent proposal produced from THSS carries that exact snapshot binding forward. A Governed State Transition (GST) then checks whether the relevant state and governance conditions still permit the write. The current public commitment API follows this bound path; the remaining limitation is provenance-sensitive enforcement across every internal review or adaptation path, as described below.

The paper makes four contributions:

- a health-specific **disclosure-conditioned, co-versioned commit contract** in which the persisted identity of the actual model-visible disclosure remains part of the later proposal’s admission context;
- a CLARA-Care reference implementation that validates snapshot identity, digest, declared evidence, state version, policy and consent versions, actor, role, purpose, task, and expiry before persistence;
- an evaluation that keeps **software-contract conformance, database concurrency and overhead, and model-context utility** separate rather than collapsing them into one superiority claim; and
- explicit negative boundaries, including a remaining commitment-level base-version-only path, bounded rather than universal concurrency evidence, false-stale costs from profile-global versioning, and the absence of independent clinical adjudication.

The evidence is versioned conservatively. Frozen artifact identifiers and null or negative findings are retained, and later reruns are reported under new implementation and run identifiers rather than retroactively attached to earlier freezes. In particular, the later 384-subject null comparison is treated as a boundary on model-utility claims rather than as a result to be hidden or explained away.

Running clinical-informatics example. Consider a medication-reconciliation task in which a patient authorizes a clinician-facing assistant to inspect the current medication list and allergy history at time t_1 . The model receives a THSS containing only the task-relevant medications, allergies, provenance, consent state, and state version. Before the model’s proposed medication correction is reviewed at t_3 , the patient withdraws the relevant sharing consent or another clinician updates the medication list at t_2 . A generic write check can ask whether the record or authority is current; GLHS additionally asks whether the proposal is still admissible when evaluated against the exact governed disclosure that the model actually used. The same pattern applies to allergy reconciliation, care-plan completion, patient-reported symptom incorporation, and other longitudinal writeback workflows.

2 Related Systems and Design Gap

The 2026 literature leaves little room for broad novelty claims around transactional memory or commit-time checks. MemTX and MemTxn already make persistent agent-memory updates transactional and evidence-aware [13, 14]. Cordon introduces a semantic transaction boundary for staged irreversible effects [25]; TOKI formalizes bitemporal write-time correctness and provenance [26]; and verified agent runtimes make read-generate-write concurrency anomalies explicit [27]. CommitGuard studies authority that becomes stale before durability [28], while Provenact formalizes policy-state serializability for concurrent agentic effects [29]. GateMem and governed shared-memory systems evaluate access control and propagation failures across multiple principals [30, 15]. Health-AI research, meanwhile, already covers bitemporal state arbitration, discrepancy reconciliation, longitudinal memory, and privacy-aware context selection [9, 11, 16]. The broader health-informatics literature also treats provenance, consent, audit metadata, and lifecycle traceability as persistent governance requirements rather than one-time preprocessing concerns [19, 20, 21, 22]. These peer-reviewed foundations motivate the health-data problem; the recent agent-memory preprints are used here as the nearest technical neighbors, not as the sole scientific basis for the novelty claim.

GLHS is therefore not positioned as a new transaction mechanism or a generic authorization system. The distinction is the *health disclosure object itself*. The system persists the purpose-, consent-, actor-, task-, state-, and evidence-bounded snapshot actually supplied to inference and requires a later proposal to name that same snapshot. CommitGuard asks whether an authority witness still licenses an effect at durability; Provenact asks whether authorization is serializable with policy state. GLHS adds a health-specific question: whether the model’s persistent proposal is still admissible *given the exact governed health information from which the proposal was derived*.

This distinction matters because read-time minimization and write-time authorization solve different problems. A system can minimize context correctly but later authorize a write without knowing which minimized disclosure the model used. It can also perform a fresh state-version check while losing the provenance, purpose, consent, or task coordinates of the inference context. GLHS binds those coordinates across the operation boundary. The contribution is therefore compositional: established temporal, provenance, authorization, and concurrency mechanisms are connected so that a concrete model-visible health snapshot remains a verifiable dependency of a later persistent effect.

3 System Model and Governance Contract

3.1 Governed longitudinal state

GLHS represents a system view that is evidence-backed and permitted for the current use, not an unquestionable biological truth. Evidence, events, assertions, state, transitions, conflicts, and projections remain separate so that provenance and uncertainty are not flattened into the current view. State can be reconstructed at clinical valid time t_v and semantic knowledge time t_k ; database write time remains in the audit trail rather than substituting for knowledge time.

For subject u , the governed state is

$$S_u(t_v, t_k) = \text{GovernedView}_u(t_v, t_k). \quad (1)$$

Assertions retain provenance, epistemic status, and lifecycle. Conflicting assertions may remain visible together until a resolution becomes both valid and known.

3.2 Task-Bounded Health State Snapshot

THSS is the governed object produced on the read path. Besides selected content, it stores the coordinates needed to reproduce what the AI was allowed to see:

$$H = \langle u, a, r, \phi, \tau, v_s, v_\pi, v_c, t_v, t_k, id_H, d_H, E_H \rangle. \quad (2)$$

Here a and r identify actor and role; ϕ and τ denote purpose and task; v_s is the state version, while v_π and v_c are separate policy and consent coordinates. We use *governance coordinates* as an umbrella term for these policy/consent values rather than implying one merged governance counter. id_H names the persisted snapshot; d_H is its consistency digest; and E_H is the disclosed content.

Canonical bytes use the project-defined `clara.canonical-json.v1` profile and SHA-256. The profile sorts object keys, preserves array order, emits compact UTF-8 JSON, normalizes timezone-aware datetimes to UTC and negative zero to zero, and rejects unsupported or non-finite values; it is not presented as RFC 8785. The persisted manifest records `glhs.snapshot.v3`, payload schema `glhs.snapshot.payload.v3`, the digest algorithm and canonicalization profile, and validation fails closed when metadata or recomputed digests disagree. Strict model-facing disclosure additionally records consent basis, expiry, assertion identifiers/hashes, evidence identifiers, and an opaque profile scope.

3.3 Snapshot-bound proposal and commit invariant

The model is not given unrestricted write access. It returns a persistent proposal

$$P = \langle u, a, r, \phi, \tau, v_s, v_\pi, v_c, id_H, d_H, \Delta, \rho \rangle, \quad (3)$$

where Δ is the proposed change and ρ records proposal provenance, including a model-manifest reference when applicable. For the snapshot-bound mode studied here,

$$\text{Commit}(P) \Leftrightarrow \text{Bound}(P, H) \wedge \text{StateCurrent}(P) \wedge \text{GovernanceCurrent}(P) \wedge \text{SnapshotValid}(P). \quad (4)$$

Bound fails when profile, actor, role, purpose, task, base state version, snapshot identity/digest, or declared evidence set disagree. At commit, the gateway obtains a **FOR UPDATE** lock on the profile row,

rechecks base state, rereads current consent, compares the frozen policy version, and writes the transition and resulting state in the same database session.

The implementation now contains a generic assertion-admission regression lock: a caller that declares it consumed THSS cannot silently fall back to the base-version-only assertion path, and a model process cannot use generic assertion admission as a write fallback. The commitment gateway, however, still exposes an explicit base-version-only proposal constructor for workflows that are not proven to have consumed THSS. Provenance is not yet strong enough to establish that every reviewed downstream commitment proposal derived from Strict THSS must remain snapshot-bound. Equation 4 therefore characterizes the stronger path that is evaluated, and the manuscript does not claim universal downgrade prevention across every review/adaptation route.

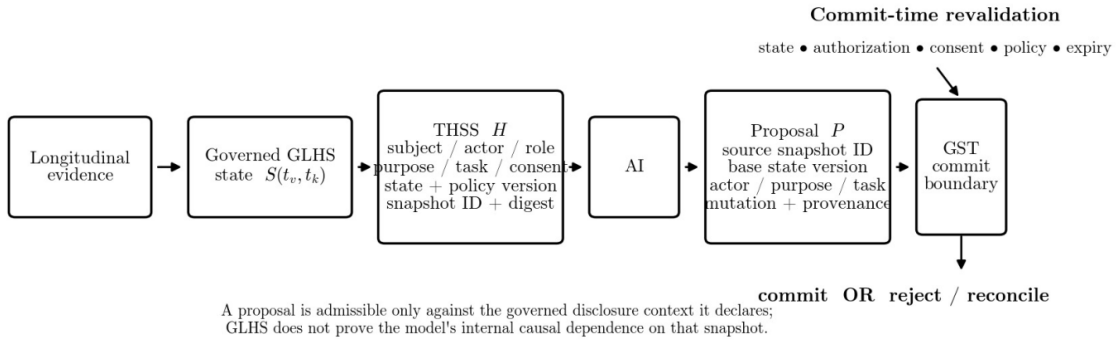


Figure 1: Core GLHS disclosure-to-commit contract. THSS constrains the model-facing disclosure at t_1 ; the record, consent, role, or policy may change at t_2 while inference/review is still in progress; the proposal carries its source coordinates; and GST at t_3 commits only after the tested binding, state-current, governance-current, and snapshot-valid checks succeed.

3.4 Failure model and claim boundary

Table 1 makes explicit which failure modes are exercised and which remain open.

Table 1: Failure modes, intended handling, and current evidence boundary.

Failure mode	Contract handling	Evidence / limitation
Stale base state	StateCurrent + profile-row lock	Exercised on PostgreSQL state-version races; stale writes rejected in tested path.
Snapshot substitution or digest mismatch	Persisted id_H , d_H and recomputation	Deterministic conformance tests fail closed on tested mismatches.
Actor/role/purpose/task mismatch	Exact coordinate equality in Bound	Exercised in clause-level tests.
Evidence-set substitution	Declared evidence membership in Bound	Exercised in deterministic conformance path.
Consent/policy change during inference	Commit-time reread/version comparison	Exercised in a frozen 12-schedule PostgreSQL TOCTOU matrix with consent, role, policy-epoch, state, and compound changes; no forbidden commit or indeterminate ordering was observed in those schedules. This does not establish correctness for arbitrary interleavings or production concurrency.
Expired disclosure	SnapshotValid	Expiry is included in revalidation/conformance tests.
Downgrade to base-version-only proposal	Requires provenance-sensitive mandatory policy	Generic assertion admission rejects declared THSS-consuming/model fallbacks, but commitment-level provenance does not yet prove mandatory snapshot binding across every reviewed downstream route.
Unrelated concurrent writes	Profile-global optimistic versioning	Fails closed but creates expected false-stale rejection; finer-grained alternatives were not run on PostgreSQL.

4 Implementation and Reproducibility Boundary

The reference implementation is part of CLARA-Care and is frozen for this manuscript at Git revision 2960acb590fda7ee914b6caded961c89f5e3b8e4. Snapshot-bound proposals are validated in the API-owned commitment gateway. Strict disclosures read persisted manifest identity and digests after canonicalization and fail if required binding fields are missing. State replay exposes separate valid-time and semantic knowledge-time cutoffs. Historical conflict visibility is reconstructed from conflict-creation and conflict-resolution transitions rather than from mutable current status. Generic and commitment THSS share the same executable five-stage trace. Each reported run keeps its own implementation SHA and checksum lineage; older results are not reassigned to this manuscript-level freeze.

Model-mediated evaluation used the completed 64-subject solver-v5 cohort on 11 August 2026. The run was tied to implementation 17dd4b8c558c2ec0cb8c2572728093d9aa3ce914, internal pre-registration SHA-256 99a6c1338af87e549fa05406c2c7248713d39fa297e56494fb5bd099efdd6bb2, and pre-execution seal 3280f9c9863a76b7bd51891f9bac28401d652d9ce714f6a340099fa37efdb456. Provider calls used an OpenAI-compatible /chat/completions endpoint with requested IDs `antigravity/claude-sonnet-4-6` and `antigravity/gemini-3.6-flash-high`; returned IDs had to match `claude-sonnet-4-6` and `gemini-3.6-flash-high`. Requests used temperature 0, `max_tokens=1024`, `stream=false`, and JSON-schema response mode; `top_p` and a provider API-version parameter were not set. Timeout was 60 s with at most two retries for declared transient HTTP failures

and 0.25-s initial backoff. The completed run had zero errors and zero retries.

Service-layer measurements were produced at implementation 36642787931e5ce429f73e8087c6a2ef66e71307 with checksum inventory 75115757d7acec428d9291c567e12a2477f909990627d3f10e8bce40a8911336. Source code and reproducibility materials are available at <https://github.com/Project-CLARA-HBT/CLARA-Care>.

5 Evaluation Design

The evidence hierarchy is intentionally contract-centric: disclosure-to-commit invariant behavior, governance/state drift handling, reconstructability, and concurrency/overhead are primary software evidence. Model-context experiments are secondary and are used to test whether bounded disclosure preserves useful task context; they are not the basis for the architectural novelty claim.

5.1 Contract clause isolation

The contract experiments and model comparators answer different questions. The standards-composed mechanism comparator includes bitemporal resolution, version-aware writes, current authorization, provenance, and audit. No single comparator was verified to match GLHS on every current consent, policy, purpose, and expiry semantic while differing only in exact snapshot binding. The 16-case matrix therefore demonstrates feature-level conformance and failure localization; it is *not* a causal estimate of the snapshot-binding contribution in isolation. A matched “full governance without exact $id_H/d_H/E_H$ dependency” versus “the same mechanism with exact binding” experiment remains a required next freeze before any binding-only causal claim.

Seven progressively stronger variants were applied to the same 16 developer-authored cases, yielding 112 deterministic conformance executions, not 112 independent samples. Variants add: temporal/provenance resolution; base-version checking; authorization at disclosure; provenance and audit; snapshot identity; full context/digest plus disclosed-evidence membership; and current reauthorization with policy, consent, and expiry checks. Separate property-based, adversarial, leakage, and boundary tests are not pooled with the 112 cells; mutation testing was not part of this study.

5.2 PostgreSQL concurrency and service overhead

State-version concurrency was exercised on PostgreSQL 16.14. A four-writer test covered both a shared semantic slot and four unrelated slots under a profile-global version counter. A second grid used 1, 2, 4, 8, and 16 writers, with five independent profile races per level, for 50 races and 310 writer attempts. Losses on the same semantic dependency were classified as true stale; rejections on unrelated slots were classified as false stale. These state-version contention runs were analyzed separately from a frozen governance-TOCTOU matrix. The latter used 12 logical PostgreSQL schedules with persisted consent-revocation, role-change, policy-epoch, state-change, and compound-drift writers. The unit of analysis was one frozen logical schedule; retries remained inside the schedule rather than becoming independent observations.

Service-layer overhead used a clean, one-worker, in-process PostgreSQL 16.14 run with read-committed isolation, synchronous commit, history depth 50, and 30 repetitions per operation. HTTP transport, worker scaling, and provider inference were excluded; p99 is omitted because 30 repetitions do not support a stable tail estimate.

5.3 Governance-writer TOCTOU and bounded assurance

The governance-TOCTOU protocol froze 12 logical schedules that interleaved proposal admission with persisted consent revocation, role change, policy-epoch advance, state advance, and compound drift. Outcomes were classified using observed ordering evidence; indeterminate orderings were never counted as safe. A separate compact finite-state model encoded the disclosure-to-commit coordinates and 11 invariants, then exhaustively enumerated reachable states to bounded depth 5. This bounded model is a finite assurance artifact rather than a proof for unbounded versions, evidence sets, actors, or arbitrary-length interleavings.

5.4 Source-disjoint utility sanity grid

A later source-disjoint synthetic structural grid used six tasks, four context conditions (Strict THSS, bound THSS, state-only, and unbound), and two locked model families for 48 calls with zero execution errors. This small grid was designed as a utility-preservation sanity check rather than a powered superiority experiment; condition-specific prompt bytes were distinct and scoring remained at the task/model level.

5.5 Frozen two-model context-utility cohort

Strict THSS was compared with full authorized history, long chronological context, Naive RAG, last-write-wins (LWW), and BTSA. These contrasts test disclosure context for model reasoning, not the later GST persistence boundary. The preregistered endpoint was subject-level all-axes exact match. Ten paired tests (five comparators for each model) were analyzed in one Holm family. Missing or malformed outputs counted as errors. Paired subject-level differences were bootstrapped 1,000 times with seed 20260811; exact two-sided sign tests used only discordant subjects and were Holm-adjusted across all ten tests. The 64-subject plan targeted at least 51 non-tied pairs and approximately 81.5% power for directional win probability 0.75 under a conservative 0.005 per-test threshold; observed significant contrasts ultimately contained only 9–21 non-tied subjects.

6 Results

6.1 Contract conformance

The focused tests exercised Eqs. 2–4. Snapshot-bound proposals were rejected for mismatches in profile, actor, role, purpose, task, base state version, snapshot identity, or digest. The write path also rechecked current policy, consent, and expiry. Incomplete binding metadata and THSS traces with missing or reordered stages were rejected. A historical regression confirmed that a conflict resolved in August still appears in a July snapshot compiled later because conflict visibility is replayed from transition history rather than read from the mutable current projection.

All 112 cells in the 16-case $\times$ 7-variant matrix produced the expected conformance result and identified the first rejecting clause. Because the standards-composed comparator is not matched on every current governance semantic, this matrix is not a causal estimate of exact snapshot binding alone.

6.2 Atomic write safety and version granularity

In the $N = 4$ PostgreSQL test, both same-slot and unrelated-slot races produced one atomic winner and three stale rejections with no database error. Under these tested conditions the profile-global contract failed closed. The contention grid reproduced the expected consequence of one winner per simultaneous profile race: unrelated-slot false-stale rejection was 0 with one writer, 0.50 with two, 0.75 with four, 0.875 with eight, and 0.9375 with sixteen. These values are a design consequence reproduced by the benchmark, not estimated population rates. Resource-partitioned and dependency-aware alternatives were not run on PostgreSQL.

6.3 Governance-writer TOCTOU and bounded assurance

Across the 12 frozen PostgreSQL governance schedules, 10 invalid post-change attempts were rejected, one transition that completed before the observed revocation served as a temporally valid control, and no forbidden commit was observed. Persisted governance writers covered consent revocation, role change, and policy-epoch advance; the analysis recorded zero indeterminate ordering and zero operational deadlock counted as safety. These 12 schedules demonstrate the enumerated interleavings only and are not evidence of universal serializability under arbitrary workloads.

The bounded depth-5 finite-state exploration reached 21,361 unique states and traversed 90,432 transitions with zero violations of 11 specified invariants. A deeper depth-6 run reached 69,342 states and 378,602 transitions with zero violations, but both runs operate over deliberately finite saturated domains and therefore are bounded assurance rather than universal proof.

6.4 Service-layer overhead

Table 2: Descriptive PostgreSQL service-layer benchmark: one worker, history depth 50, 30 repetitions per operation. Values include transaction finalization but exclude HTTP and model inference.

Operation	p50 ms	p95 ms	Sequential ops/s
GST transition	30.967	54.685	29.952
State reconstruction	7.298	12.299	129.688
THSS compilation	61.806	68.889	16.937
Governed-decision reconstruction	10.537	17.011	95.350
Audit lookup	2.122	5.501	399.149
Invalidate + rebuild	52.996	74.336	18.295
Enter-in-error + rebuild	69.314	87.309	13.818

Peak process RSS was approximately 109 MiB. These measurements describe one local, in-process service-layer environment and are not public-HTTP latency, deployment capacity, or production throughput.

6.5 Prospective two-model evaluation

The solver-v5 benchmark completed all 1,152 model-condition cells with zero errors and zero retries. Across cells, accuracy was 93.06% for lifecycle, 96.79% for evidence, 99.22% for timeliness, and 99.83% for escalation; 1,027/1,152 cells (89.15%) were exact on all four axes. On the subject-level primary endpoint, Strict THSS reached 63/64 (98.44%) with Claude and 64/64 (100%) with Gemini. The single

remaining Strict-THSS Claude error—a cancelled history predicted as satisfied—was retained without post-hoc repair. The 128 construction-review requests served a separate bounded purpose: two model reviews per subject checked deterministic anchor construction while code retained final authority. All 64 constructions were accepted; candidate-slot F1 and due-window exact accuracy were both 1.0.

Table 3: Ten preregistered subject-level contrasts for Strict THSS. Positive effects favor Strict THSS; Holm adjustment spans the complete ten-test family.

Model	Comparator	Effect (pp)	95% bootstrap CI	Holm result	Discordance
Claude	Naive RAG	+32.81	+21.88 to +43.75	$p = 0.00000954$	21
Claude	LWW	+25.00	+15.63 to +35.94	$p = 0.00027466$	16
Claude	Full history	+18.75	+7.81 to +29.69	$p = 0.01098633$	14
Claude	Long context	+14.06	+6.25 to +23.44	$p = 0.01953125$	9
Claude	BTSA	+4.69	not reported	$p = 1.0$, n.s.	not reported
Gemini	Naive RAG	+25.00	+14.06 to +35.94	$p = 0.00027466$	16
Gemini	LWW	+25.00	+15.63 to +37.50	$p = 0.00027466$	16
Gemini	BTSA	0.00	exact tie	n.s.	exact tie
Gemini	Full history	0.00	exact tie	n.s.	exact tie
Gemini	Long context	0.00	exact tie	n.s.	exact tie

Six contrasts remained Holm-significant, but only 9–21 discordant subjects informed those tests, below the planned 51 informative pairs. Claude versus BTSA was not significant, and Gemini tied BTSA, full history, and long context exactly. The exact sign tests remain valid for observed discordances, but the high tie rate limits effect-size precision and confirmatory strength.

6.6 Later synthetic checks and a null comparator result

A later sealed Claude-only synthetic run used 384 independent subjects, nine context conditions, and 3,456 solver cells. The prespecified subject-level contrast between Strict THSS and full authorized history was null: 70 wins, 73 losses, and 241 ties; mean Strict-minus-full-history effect -0.0078125 ; 95% bootstrap CI $[-0.0677083, 0.0520833]$; exact two-sided sign-test $p = 0.8672499071$. We therefore do not claim universal model-level superiority of Strict THSS over complete authorized history. The run also contained 220 fail-closed malformed outputs and is reported as descriptive synthetic software evidence rather than clinical validation.

In the subsequent six-task source-disjoint utility grid, Strict THSS was correct on 11/12 task-model cells (91.7%), bound THSS on 12/12 (100%), state-only on 11/12 (91.7%), and unbound context on 9/12 (75.0%). Gemini 3.6 Flash High was correct on 23/24 cells (95.8%) and Claude Sonnet 4.6 on 20/24 (83.3%). Because the grid contains only six tasks and 12 cells per condition, these values are a small sanity check against catastrophic utility loss, not a powered noninferiority or superiority result.

7 Discussion

7.1 What the contract adds

The strongest result is not that GLHS makes language models uniformly more accurate. The later 384-subject comparison does not support that claim. The more defensible contribution is architectural: under the tested conditions, the system can preserve the exact governed context shown to a model and carry that context into the later write-admission decision. This makes it possible to reconstruct not only

what was written, but also which governed disclosure the proposal depended on and which state, policy, consent, actor, purpose, and task coordinates were checked before persistence.

This changes the failure model from “the model saw some authorized data” to a stronger and more testable statement: “this proposal was derived from this exact governed disclosure, and that disclosure was still admissible for this write when the system attempted to persist it.” The contract does not remove the need for ordinary authorization, concurrency control, provenance, or human review. It gives those mechanisms a shared object to validate across the read–write interval.

7.2 Relation to governed and transactional agent memory

Transactional and governed-memory systems provide important neighboring mechanisms, including source-supported updates, recovery, scoped propagation, and commit-time checks [13, 14, 15, 28, 29]. GLHS should be read as a health-specific composition of these concerns rather than as a replacement for them. Its distinctive object is the task-bounded health disclosure actually consumed by inference, together with its governance coordinates and provenance. The current evaluation supports this composition only within the frozen software and synthetic experiments reported here.

7.3 Privacy, safety, and operational trade-offs

THSS treats PII, PHI, and other sensitive attributes as governed data, not as ordinary prompt context. The compiler can omit, minimize, or pseudonymize information according to the actor, purpose, task, consent state, and policy while retaining provenance in the governed record. Evaluation artifacts use synthetic or de-identified inputs where applicable, opaque study-local subject tokens, and content hashes. These controls improve traceability but do not by themselves establish legal “minimum necessary” compliance, anonymization, sender authentication, or a general privacy guarantee.

The concurrency results also expose a practical trade-off. A profile-global version counter is easy to reason about, but unrelated writes compete for the same version. At 16 writers, the unrelated-slot false-stale rate reached 0.9375. Finer-grained versions could reduce unnecessary rejection, but splitting the version boundary would require new safety evidence because lost updates and cross-dependency errors can reappear if related state is no longer checked together.

8 Limitations and Versioning Roadmap

The model-context evidence remains synthetic. The prospective cohort contains 64 subjects, and frequent ties left only 9–21 informative pairs in the significant comparisons rather than the planned 51. Strict THSS was not significantly better than BTSA for Claude, while Gemini tied BTSA, full history, and long context. No independent human clinical adjudication or real-world clinical validation is reported.

The implementation also has important systems limitations. Generic assertion admission now rejects declared THSS-consuming/model fallbacks to an unbound write, but the commitment gateway retains an explicit base-version-only constructor and current provenance does not prove that every reviewed downstream proposal consumed Strict THSS. Mandatory snapshot retention is therefore not established across every commitment-review path. Governance-writer behavior has been exercised in a frozen 12-schedule PostgreSQL matrix with persisted consent, role, policy-epoch, and state writers, but this does not establish correctness for arbitrary interleavings, higher concurrency, or production EHR deployments. No comparator is known to match every GLHS governance semantic while differing only in exact snapshot

binding. The service experiments use a single in-process worker, and the current HTTP/cache/retrieval path has not been subjected to a fully deployed adversarial matrix.

The model-utility evidence also argues for restraint. The later 384-subject comparison between Strict THSS and full authorized history was null, and the newest source-disjoint utility grid contains only six tasks. These findings are retained because they constrain the claim: GLHS is currently supported as a governance and traceability mechanism, not as a universally superior context-selection strategy.

A stronger future version should report newly frozen evidence only after the corresponding implementation changes are complete. Priority items include mandatory provenance-sensitive snapshot binding for model-derived proposals, a matched comparator that differs only in binding semantics, independent adjudication, broader governance and concurrency scaling, and finer-grained versioning if it can be introduced without weakening safety. Earlier artifacts and hashes should remain available rather than being silently relabeled.

9 Conclusion

GLHS treats persistent longitudinal health AI as a governed read–write process rather than a sequence of unrelated retrieval and update steps. THSS records the exact state and governance context supplied to the model, a snapshot-bound proposal carries that identity forward, and GST checks whether the proposal is still admissible before persistence. In the tested software paths, the implementation rejected the specified context mismatches, prevented stale commits in the state-version races, and observed no forbidden commit in the frozen 12-schedule governance-TOCTOU matrix. Bounded exploration found no violation of 11 invariants across 21,361 states and 90,432 transitions at depth 5.

The model-context results are mixed in a scientifically useful way. Strict THSS reached 98.44% all-axes exact match with Claude and 100% with Gemini in the frozen 64-subject cohort, with six prespecified Holm-significant comparisons but fewer informative pairs than planned. A later 384-subject comparison with full authorized history was null. Taken together, these results support a traceable software path from governed disclosure to persistent write under the tested conditions, while arguing against claims of universal model-level superiority. They do not establish clinical benefit, universal downgrade resistance, correctness for arbitrary unbounded interleavings, or regulatory compliance.

Statements and Declarations

Competing interests. The authors have no relevant financial or non-financial interests to disclose.

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Ethical approval. Not applicable. The experiments were software evaluations conducted on synthetic data and did not involve human participants, human biological material, or identifiable patient data.

Consent to participate. Not applicable. No human participants were recruited.

Consent to publish. Not applicable. The manuscript contains no identifiable information from individual participants.

Data and code availability. Source code and reproducibility materials are publicly available in the CLARA-Care repository: <https://github.com/Project-CLARA-HBT/CLARA-Care>. Run-specific identifiers and source revisions are reported with the corresponding experiments so that results from different implementation freezes are not conflated.

Use of generative AI. Generative AI tools were used for language editing and literature-search assistance. They did not generate the reported experimental data. The authors verified the cited sources, numerical results, analyses, and scientific claims and remain responsible for the manuscript.

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